

Unit 13, Lesson 2: Solving for Unknown Angles

Benchmark

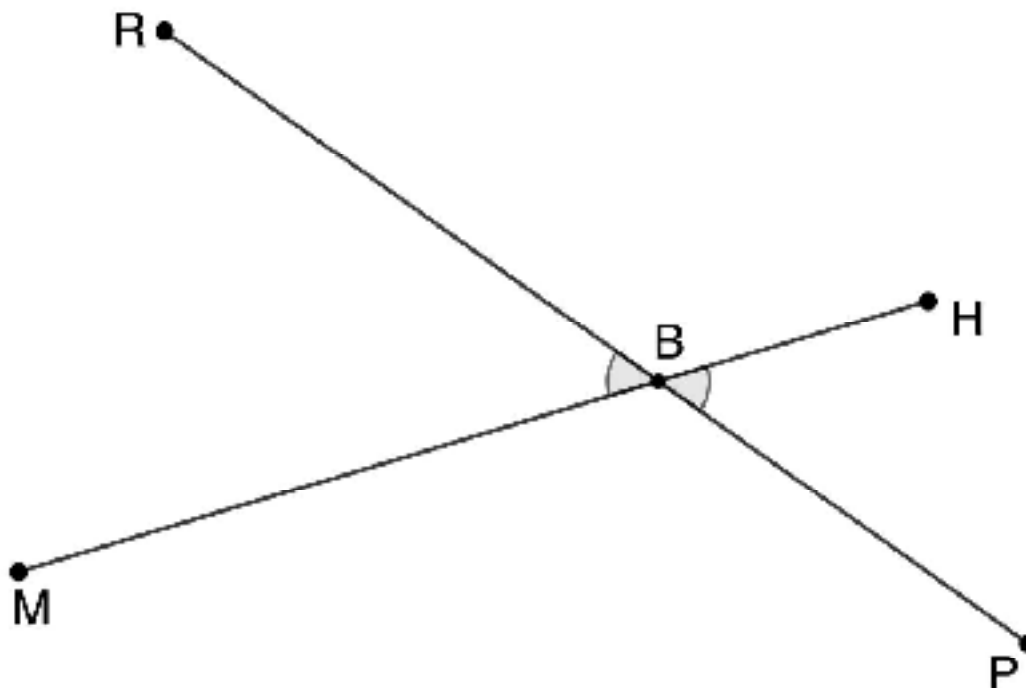
MA.B.GR.1.4 Solve mathematical problems involving the relationships between supplementary, complementary, vertical or adjacent angles.

Learning Targets

- ☐ I can apply angle relationships to find missing angle measures.
- ☐ I can write and solve algebraic equations to solve for unknown angles.

13.2.1 Warm-Up: Agree or Disagree

1. Line segments MR and HP intersect at point B .



- a. Jamessia says $\angle MBR$ and $\angle PBH$ have the same measure. Malik disagrees and says that obviously the measure of $\angle MBR$ is larger.

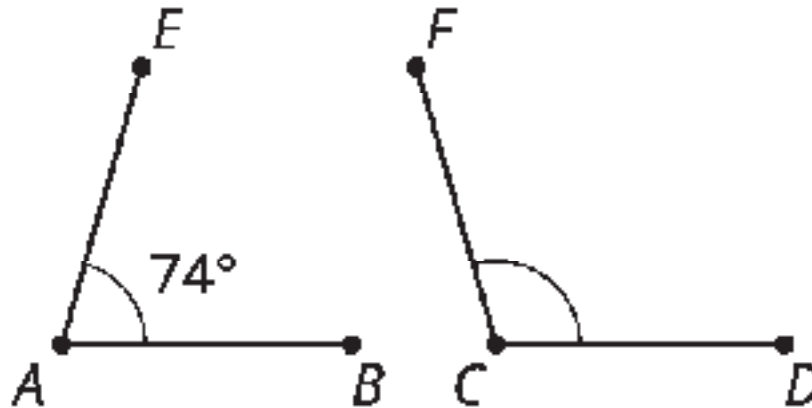
Where do you stand?

- ☐ I agree with Jamessia.
 - ☐ I agree with Malik.
 - ☐ I disagree with both students.
- b. Explain your choice for Part A.

13.2.2 Guided Instruction: Applying Relationships to Find Missing Angles

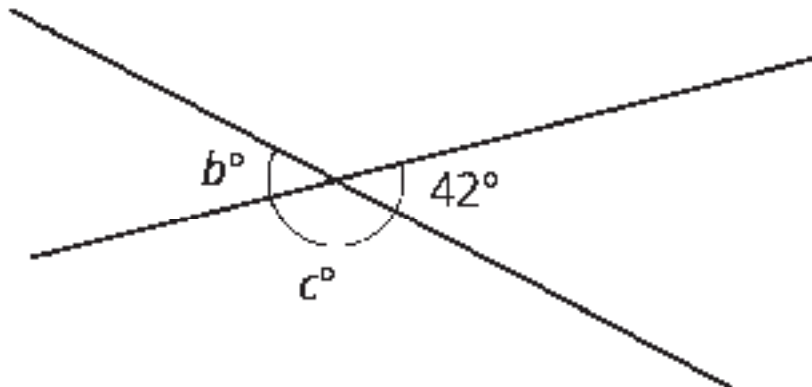
Angle relationships can be used to solve mathematical problems and find missing angles.

1. Angles EAB and FCD are supplementary, and the measure of $\angle EAB$ is 74° .



Find the measure of angle FCD .

2. Two line segments intersect, as shown.



- a. Discuss with your partner the relationships between the angles marked in the diagram and how those relationships can be used to determine the missing angle measures.

b. Complete the statements.

The value of b is

- ☐ 42°
- ☐ 48°
- ☐ 138°

because

- ☐ complementary
- ☐ supplementary
- ☐ vertical

angles

have

- ☐ the same measure.
- ☐ a sum of 90° .
- ☐ a sum of 180° .

The value of c is

- ☐ 42°
- ☐ 48°
- ☐ 138°

because

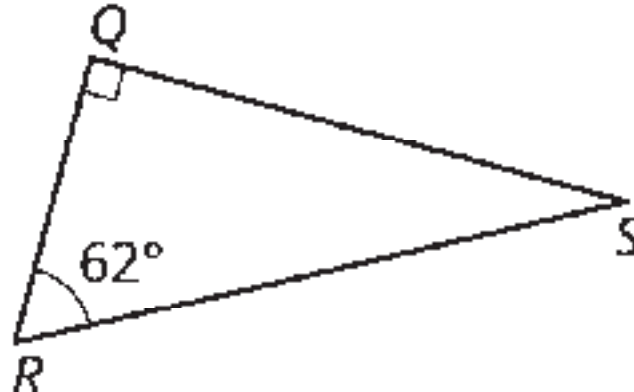
- ☐ complementary
- ☐ supplementary
- ☐ vertical

angles

have

- ☐ the same measure.
- ☐ a sum of 90° .
- ☐ a sum of 180° .

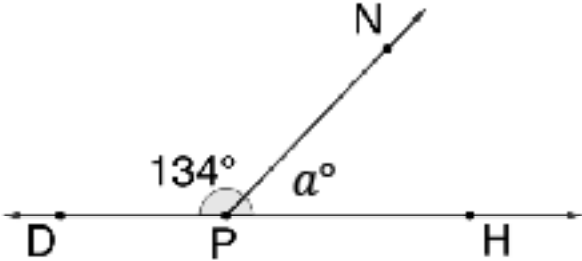
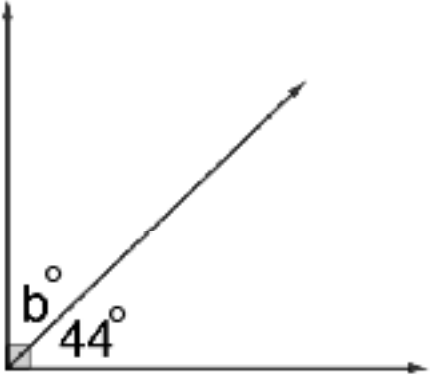
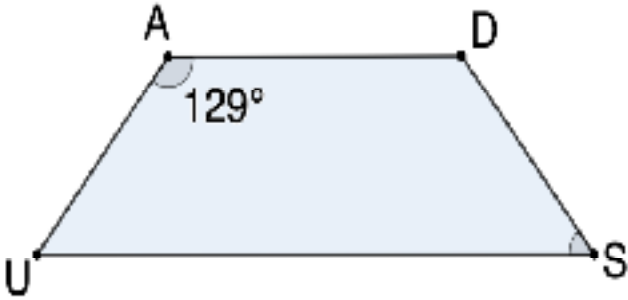
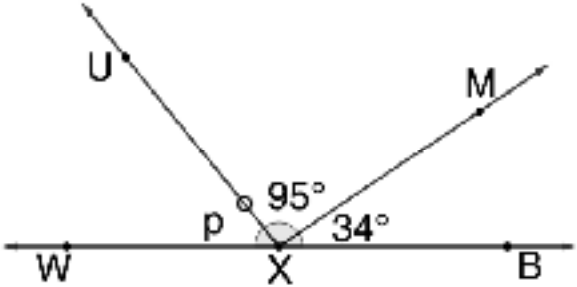
3. In the figure, angles QRS and RSQ are complementary.



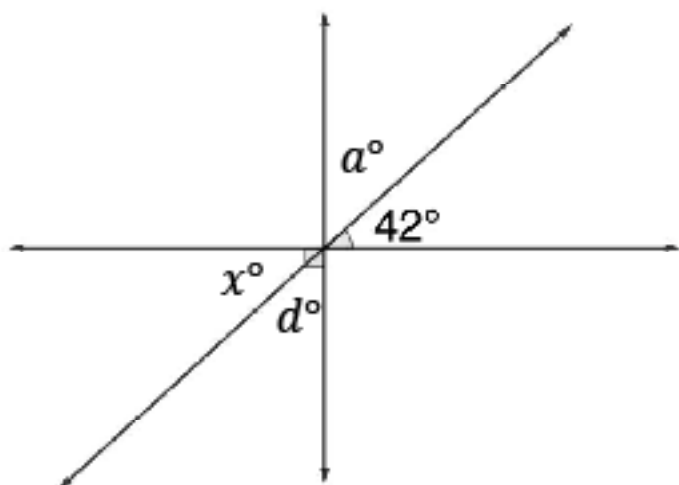
Find the measure of angle RSQ .

13.2.3 Collaboration: Applying Relationships to Find Missing Angles

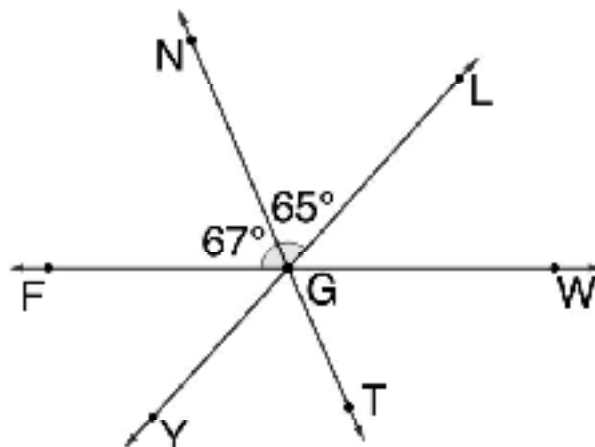
1. With your partner, determine who will be Partner A and who will be Partner B.
 - a. Complete the problems in the appropriate column independently.

Partner A	Partner B
Find the value of a. 	Find the value of b. 
Angle DAU and angle USD are supplementary. Find the measure of angle USD. 	Find the value of p. 

Find the value of d .



Find the measure of $\angle LGW$.



- b. The problems in each row have the same result. Check your work with your partner. If there are discrepancies in your answers, work together to find and correct any errors.

2. Three figures are shown with angles measuring g° and h° marked in each figure.

Figure A

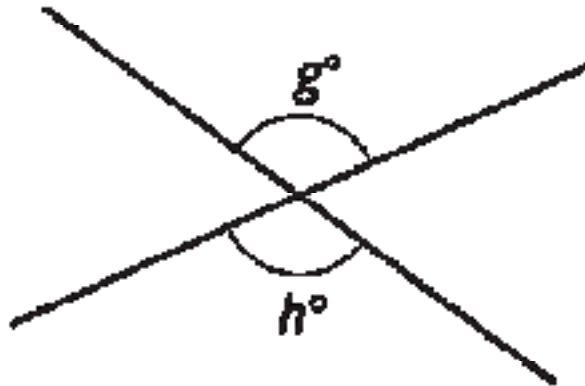


Figure B

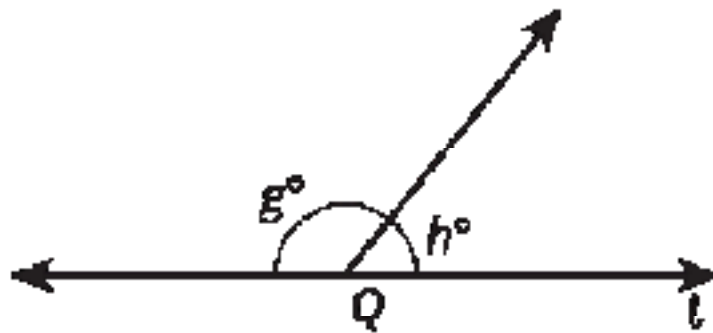
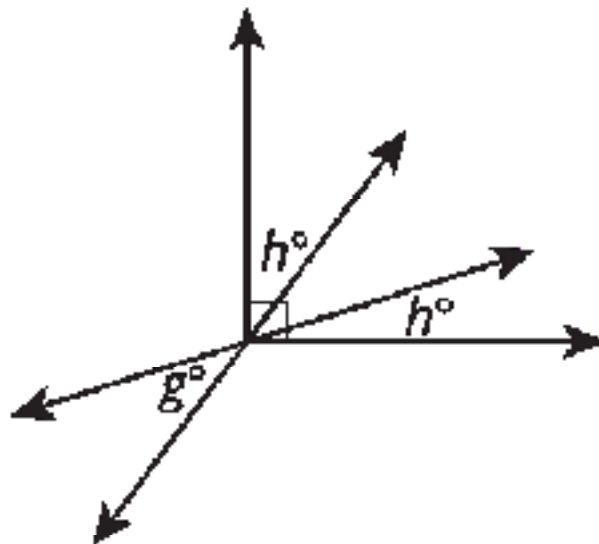


Figure C



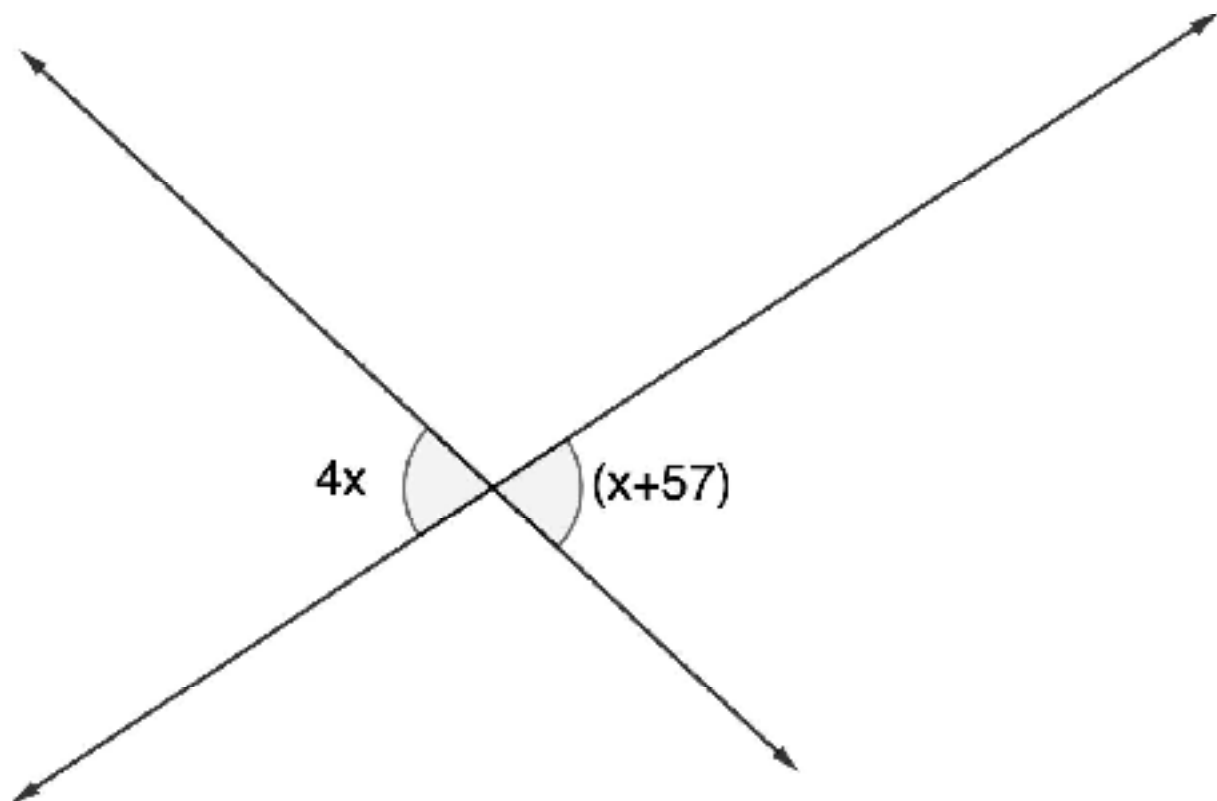
Work with your partner to complete the table by matching each figure to the equation that represents the relationships between the angles in the figure. For each match, explain how you know they are a match.

Equation	Figure	Explanation
$g + h = 180$		
$g = h$		
$2h + g = 90$		

13.2.4 Guided Practice: Writing Equations to Find Missing Angles

Algebraic equations can be written to express the relationship between complementary, supplementary, and vertical angles. The equations can then be solved to find the measures of unknown angles.

1. Two angles formed by intersecting lines are labeled.



- a. The angles marked are

- ☐ complementary angles,
- ☐ supplementary angles,
- ☐ vertical angles,

so they have

- ☐ the same measure.
- ☐ a sum of 90° .
- ☐ a sum of 180° .

- b. The equation

- ☐ $4x + x + 57 = 180$
- ☐ $4x + x + 57 = 90$
- ☐ $4x = x + 57$

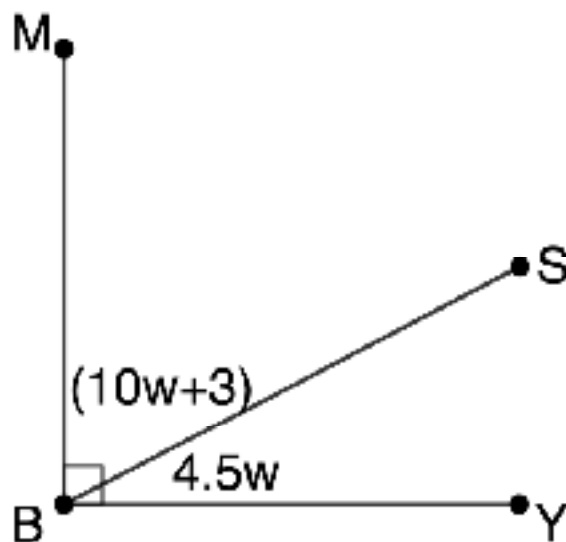
can be used to

find the value of x .

c. Solve for x .

d. Find the measure of each angle using the solution to Part C.

2. Angle MBY is a right angle.



a. Write an equation that can be used to find the value of w .

b. **Solve the equation.**

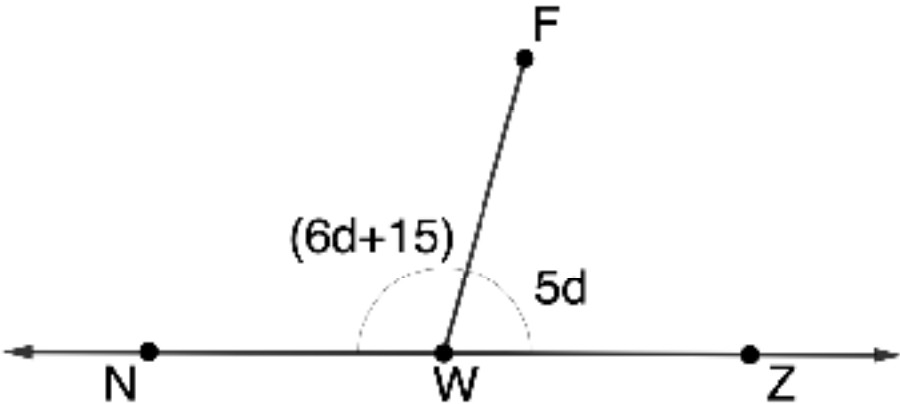
c. **Use the solution to find the measure of each angle.**

Angle	Measure
<i>$\angle MBS$</i>	
<i>$\angle SBY$</i>	

d. **Explain whether the angle measures determined in Part C make sense based on the relationship between $\angle MBS$ and $\angle SBY$.**

13.2.5 Wrap-Up: Ticket Out the Door

1. Two angles are labeled in the diagram of line NZ , which intersects with line segment WF .



- a. Write an equation that can be used to find the value of d .
- b. Solve the equation.
- c. Use the solution to find the measure of each angle.

Angle	Measure
$\angle FWN$	
$\angle FWZ$	